$$\begin{pmatrix}1 & 40 & 10\\ 1 & 35 & 9\\ 1 & 40 & 12\end{pmatrix}\begin{pmatrix}a\\ b\\ c\end{pmatrix}=\begin{pmatrix}6950\\ 6725\\ 7100\end{pmatrix}$$

$$A \quad X = B$$

Augmented matrix $[A,B]$	Elementary Transformation
$\begin{pmatrix}1 & 40 & 10 & 6950\\ 1 & 35 & 9 & 6725\\ 1 & 40 & 12 & 7100\end{pmatrix}$	
$\sim\begin{pmatrix}1 & 40 & 10 & 6950\\ 0 & -5 & -1 & -225\\ 0 & 0 & 2 & 150\end{pmatrix}$	$R_2 \to R_2 - R_1$ $R_3 \to R_3 - R_1$
$\rho(A)=3, \rho([A,B])=3$	

∴ The given system is equivalent to the matrix equation

$$\begin{pmatrix}1 & 40 & 10\\ 0 & -5 & -1\\ 0 & 0 & 2\end{pmatrix}\begin{pmatrix}a\\ b\\ c\end{pmatrix}=\begin{pmatrix}6950\\ -225\\ 150\end{pmatrix}$$

$$a+40b+10c=6950 \quad (1)$$

$$-5b-c=-225 \quad (2)$$

$$2c=150 \quad (3)$$

$$\boxed{c=75}$$

Now, $(2) \Rightarrow -5b-75=-225$

$$\boxed{b=30}$$

and $(1) \Rightarrow a+1200+750=6950$

$$a=5000$$

$$a=5000,\ b=30,\ c=75$$

∴ The production equation is

$$P = 5000+30l+75m$$

$$P_{\text{at } l=50,\ m=15} = 5000+30(50)+75(15)$$

$$= 7625$$

∴ The production = 7,625 units.

Exercise 1.1

1. Find the rank of each of the following matrices

i) $\begin{pmatrix}5 & 6\\ 7 & 8\end{pmatrix}$ ii) $\begin{pmatrix}1 & -1\\ 3 & -6\end{pmatrix}$

iii) $\begin{pmatrix}1 & 4\\ 2 & 8\end{pmatrix}$ iv) $\begin{pmatrix}2 & -1 & 1\\ 3 & 1 & -5\\ 1 & 1 & 1\end{pmatrix}$

v) $\begin{pmatrix}-1 & 2 & -2\\ 4 & -3 & 4\\ -2 & 4 & -4\end{pmatrix}$ vi) $\begin{pmatrix}1 & 2 & -1 & 3\\ 2 & 4 & 1 & -2\\ 3 & 6 & 3 & -7\end{pmatrix}$

vii) $\begin{pmatrix}3 & 1 & -5 & -1\\ 1 & -2 & 1 & -5\\ 1 & 5 & -7 & 2\end{pmatrix}$

viii) $\begin{pmatrix}1 & -2 & 3 & 4\\ -2 & 4 & -1 & -3\\ -1 & 2 & 7 & 6\end{pmatrix}$

2. If $A=\begin{pmatrix}1 & 1 & -1\\ 2 & -3 & 4\\ 3 & -2 & 3\end{pmatrix}$ *and* $B=\begin{pmatrix}1 & -2 & 3\\ -2 & 4 & -6\\ 5 & 1 & -1\end{pmatrix}$, then find the rank of AB and the rank of BA.

3. Solve the following system of equations by rank method: $x+y+z=9$, $2x+5y+7z=52$, $2x-y-z=0$

4. Show that the equations $5x+3y+7z=4$, $3x+26y+2z=9$, $7x+2y+10z=5$ are consistent and solve them by rank method.

5. Show that the following system of equations have unique solution: $x+y+z=3$, $x+2y+3z=4$, $x+4y+9z=6$ by rank method.

6. For what values of the parameter λ, will the following equations fail to have unique solution: $3x-y+\lambda z=1$, $2x+y+z=2$, $x+2y-\lambda z=-1$ by rank method.

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